

Mean Shift Clustering

A Non-Parametric Clustering Algorithm

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Outline

- 1 Introduction & Intuition
- 2 Mathematical Foundation
- 3 Advantages & Limitations
- 4 Conclusion

Historical Context

The paper about Mean Shift was first introduced by **Fukunaga and Hostetler in 1975** with the title "*The Estimation of the Gradient of a Density Function, with Applications in Pattern Recognition*" where they presented the mean shift algorithm as a method for finding modes of a density function given discrete data sampled from that function.

Later on in 2002, **Comaniciu and Meer** popularized the algorithm in their paper "*Mean Shift: A Robust Approach Toward Feature Space Analysis*" by demonstrating its effectiveness in computer vision tasks such as image segmentation and tracking.

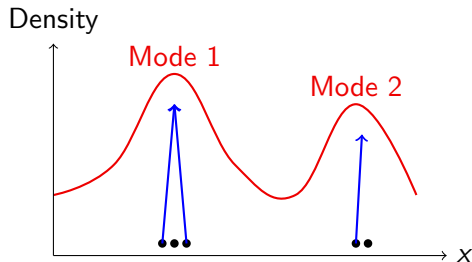
What is Mean Shift Clustering?

Core Idea:

- Find modes (peaks) of data density
- Points converge to nearest mode
- No need to specify number of clusters
- Density-based approach

Key Characteristic:

- **Non-parametric**
- Discovers clusters naturally
- Based on kernel density estimation

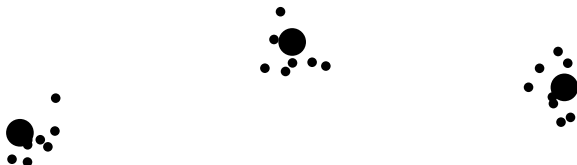


Points shift toward density peaks

Visual Intuition

Think of it as:

- 1 Imagine data points as balls on a hilly landscape
- 2 Each ball rolls uphill to the nearest peak
- 3 Balls ending at the same peak = same cluster



○ Data points Cluster modes

Advantage: Discovers cluster count automatically from data!

The Mathematics (Brief)

1. Kernel Density Estimation: Given n data points $\{x_1, x_2, \dots, x_n\}$, the kernel density estimate at point x is:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

where we use a **symmetric kernel**: $K(x) = c_k \cdot k(x^2)$

- $k : \mathbb{R}^+ \rightarrow \mathbb{R}$ is the **kernel profile** (function of squared distance)
- $h > 0$ is the bandwidth parameter
- c_k is a normalization constant
- x_i are the data points and no need to be inside the bandwidth

Key insight: As h increases, more points contribute to the density estimate, which may cause performance issues. Also if we set h too large, we may suffer from over-smoothing.

From KDE to Mean Shift

2. Deriving the Mean Shift Vector:

To find density modes, we take the **gradient** of $\hat{f}(x)$ and set it to zero.

Define the **shadow kernel profile**: $g(s) = -k'(s)$ (negative derivative of k)

Taking $\nabla \hat{f}(x) = 0$ and rearranging yields:

$$x = \frac{\sum_{i=1}^n x_i g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}{\sum_{i=1}^n g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}$$

Key insight: The function $g = -k'$ ensures we move **against** the gradient direction (toward increasing density). This weighted mean tells us where the mode is!

The Mathematics (Contd.)

3. Mean Shift Vector:

The difference between the weighted mean and current position:

$$m(x) = \underbrace{\frac{\sum_{i=1}^n x_i g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}{\sum_{i=1}^n g\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}}_{\text{weighted mean using } g} - x$$

4. Update Rule:

$$x^{(t+1)} = x^{(t)} + m(x^{(t)})$$

Key insight: Since $g = -k'$, the mean shift vector always points toward increasing density (gradient ascent on the density surface).

Common Kernel Profiles

Let $s = \left\| \frac{x - x_i}{h} \right\|^2$ be the squared normalized distance.

Gaussian:

$$k(s) = \exp\left(-\frac{s}{2}\right)$$
$$g(s) = -k'(s) = \frac{1}{2} \exp\left(-\frac{s}{2}\right)$$

Epanechnikov:

$$k(s) = \begin{cases} 1 - s & s \leq 1 \\ 0 & \text{else} \end{cases}$$
$$g(s) = \begin{cases} 1 & s \leq 1 \\ 0 & \text{else} \end{cases}$$

Flat (Uniform):

$$k(s) = \begin{cases} 1 & s \leq 1 \\ 0 & \text{else} \end{cases}$$
$$g(s) = 0 \text{ (undefined at boundary)}$$

Why Epanechnikov?

- $g(s) = 1$ (constant) $\Rightarrow$ uniform weights
- Simple weighted mean of neighbors
- Most efficient computation
- Used by sklearn's MeanShift

Why not Flat?

Algorithm Steps

Algorithm 1 Mean Shift Clustering

- 1: **Input:** Data points $\{x_1, \dots, x_n\}$, bandwidth h , kernel profile k , ϵ for convergence
- 2: **Output:** Cluster assignments
- 3:
- 4: **for** each data point x_i **do**
- 5: $y \leftarrow x_i$ ▷ Initialize at data point
- 6: **while** not converged **do**
- 7: Compute mean shift: $m(y)$
- 8: Update: $y \leftarrow y + m(y)$
- 9: **end while**
- 10: Store converged point (mode) for x_i
- 11: **end for**
- 12:
- 13: Group points by their converged modes
- 14: **Return** cluster assignments

Advantages

① No predefined cluster count

- ▶ Automatically discovers number of clusters
- ▶ Unlike K-means which requires k

② Flexible cluster shapes

- ▶ Can find non-spherical clusters
- ▶ Follows natural density contours

③ Single parameter: bandwidth h

- ▶ Controls granularity of clustering
- ▶ Easier to tune than multiple parameters

④ Robust to initialization

- ▶ Deterministic convergence to modes
- ▶ No random initialization sensitivity

Limitations (and Why They Matter)

① Computational complexity: $O(n^2 \cdot T)$ (T is the avg number of iterations to converge)

Why? Each point computes distance to all others at each iteration

Impact: Slow for large datasets ($n > 10,000$)

Mitigation: Use spatial indexing (KD-trees), binning, or sampling

② Bandwidth selection is critical

Why? Too small $h \rightarrow$ many tiny clusters; too large $h \rightarrow$ all points in one cluster

Impact: Results highly sensitive to h

Mitigation: Cross-validation, adaptive bandwidth, or domain knowledge

③ Curse of dimensionality

Why? Density estimation becomes unreliable in high dimensions

Impact: Poor performance for $d > 10$

Mitigation: Dimensionality reduction (PCA, t-SNE) first

When to Use Mean Shift?

Good Scenarios:

- Unknown number of clusters
- Non-spherical cluster shapes
- Small to medium datasets
- Image segmentation
- Object tracking in video
- Mode-seeking tasks

Avoid When:

- Very large datasets ($n > 50,000$)
- High-dimensional data ($d > 10$)
- Real-time processing needed
- Clusters have similar densities
- Well-separated spherical clusters (use K-means instead)

Summary

Mean Shift Clustering:

- **Core idea:** Find density modes, points converge to nearest mode
- **Strengths:** No cluster count needed, flexible shapes, robust
- **Weaknesses:** $O(n^2)$ complexity, bandwidth sensitivity, high-dim issues

Experimental Findings:

- Outperforms K-means in cluster quality (synthetic & Iris)
- Requires dimensionality reduction for high-dim data (Wine)
- Excellent for image segmentation, but slower
- Bandwidth selection is critical

Best Use Cases:

- Small-medium datasets with unknown cluster count
- Non-spherical clusters or mode-finding tasks
- Image/video processing applications

References

- ① Fukunaga, K., & Hostetler, L. (1975). *The estimation of the gradient of a density function, with applications in pattern recognition*. IEEE Transactions on Information Theory.
- ② Comaniciu, D., & Meer, P. (2002). *Mean shift: A robust approach toward feature space analysis*. IEEE Transactions on Pattern Analysis and Machine Intelligence.
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Thank You!

Questions?

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